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Change3D: Revisiting Change Detection and Captioning from A Video Modeling Perspective

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https://zhuduowang.github.io/Change3D

Abstract

In this paper, we present Change3D, a framework that reconceptualizes the change detection and captioning tasks through video modeling. Recent methods have achieved remarkable success by regarding each pair of bi-temporal images as separate frames. They employ a shared-weight image encoder to extract spatial features and then use a change extractor to capture differences between the two images. However, image feature encoding, being a task-agnostic process, cannot attend to changed regions effectively. Furthermore, different change extractors designed for various change detection and captioning tasks make it difficult to have a unified framework. To tackle these challenges, Change3D regards the bi-temporal images as comprising two frames akin to a tiny video. By integrating learnable perception frames between the bi-temporal images, a video encoder enables the perception frames to interact with the images directly and perceive their differences. Therefore, we can get rid of the intricate change extractors, providing a unified framework for different change detection and captioning tasks. We verify Change3D on multiple tasks, encompassing change detection (including binary change detection, semantic change detection, and building damage assessment) and change captioning, across eight standard benchmarks. Without bells and whistles, this simple yet effective framework can achieve superior performance with an ultra-light video model comprising only ∼6%-13% of the parameters and ∼8%-34% of the FLOPs compared to state-of-the-art methods. We hope that Change3D could be an alternative to 2D-based models and facilitate future research.

1. Introduction

Change detection and captioning play a crucial role in the remote sensing field, employing pairs of bi-temporal images taken from the same geographic area at different times to track and analyze changes on the Earth's surface over time [63]. It has been widely applied in various applications such as damage assessment [8, 86], urban planning [74], arable land protection [55], and environmental management [38]. Existing studies have explored various 2D models for bi-temporal change detection and captioning modeling, such as Convolutional Neural Networks (CNNs)-based [24, 79, 85, 88], Transformer-based [1, 8, 48, 83], and their hybrid models [7, 50, 88]. As shown in Fig. 2 (a), most of the methods typically follow such a paradigm: (1) A pair of bi-temporal images are treated as distinct inputs, processing each image individually through a shared-weight image encoder (*e.g.*, ResNet [29], UNet [64], Swin Transformer [52]) for spatial feature extraction. (2) Then, a well-designed change extractor, primarily leveraging attention mechanisms [70] like cross-attention [5] or spatial-channel attention [78], is introduced for bi-temporal feature interaction to detect the differences between them. (3) Finally, a decoder is employed to restore the feature map to its original resolution.

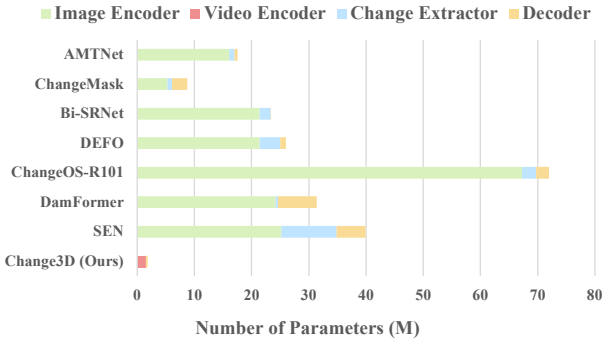


Figure 1. The parameter distribution in existing change detection and captioning methods indicates most parameters focused on image encoding, with few allocated to change extraction. This imbalance suggests an insufficient emphasis on task-related parameter learning. In contrast, our approach primarily focuses on video encoding, a task-specific process that effectively extracts changes.

While encouraging results have been observed, these

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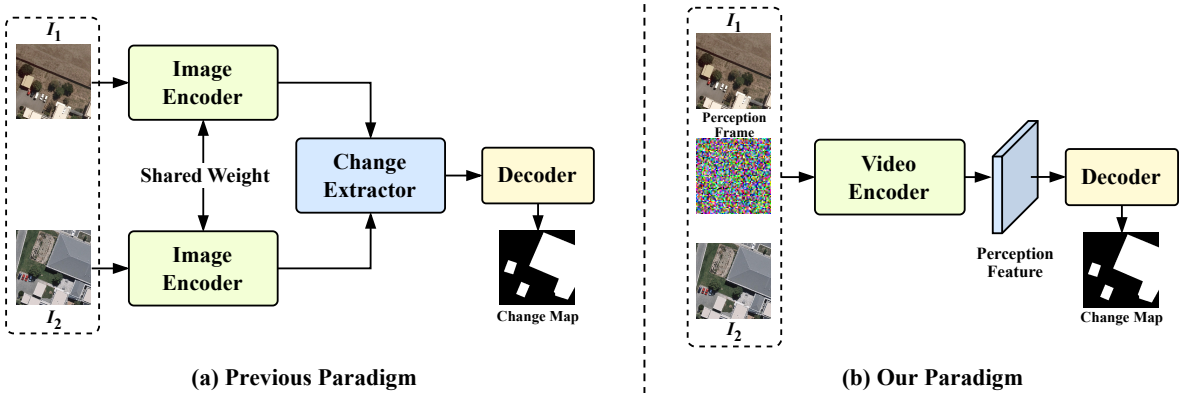


Figure 2. **Previous paradigm *vs.* our paradigm.** (a) Previous paradigm treats bi-temporal image pairs as separate inputs, where each is processed individually by a shared-weight image encoder to extract spatial features, followed by a dedicated change extractor to capture differences and a decoder to make predictions. (b) Our proposed paradigm rethinks the change detection and captioning tasks from a video modeling perspective. By incorporating a learnable perception frame between the bi-temporal images, a video encoder facilitates direct interaction between the perception frame and images to extract differences, eliminating the need for intricate change extractors and providing a unified framework for multiple tasks.

methods have the following issues: (1) The image encoder is designed to learn spatial features independently for bi-temporal images, lacking task-specificity in modeling changes. Although a previous method [20] introduces inter-frame feature interaction during image encoding, it cannot extract changes and still needs a dedicated change extractor. Moreover, as shown in Fig. 1, this paradigm results in an unreasonable parameter distribution, with the majority focusing on independent image encoding and few dedicating to change extraction, without emphasizing the parameter learning for change detection and captioning tasks. (2) Current methods necessitate separate design considerations for change extractors in different tasks, hindering the utilization of a unified framework for addressing all change detection and captioning tasks effectively. Inspired by video modeling approaches [21, 22, 40] that they can effectively model the relations between images, we seek to revisit the bi-temporal images as a tiny video, leveraging the spatiotemporal learning capability inherent in video models to capture changes.

To this end, we propose a unified framework named **Change3D**, serving as a competitive alternative to 2D-model-based approaches. Our key idea is to rethink the extraction of bi-temporal changes through the lens of video modeling techniques. As depicted in Fig. 2 (b), we ingeniously redefine bi-temporal change detection and captioning as a task akin to video modeling by incorporating a learnable frame that can learn to perceive. In particular, we create a 3D volume by aligning bi-temporal images with the learnable perception frame, all arranged along the time dimension. Change3D then utilizes a video encoder to process this input, enabling the perception frame to observe and discern the differences between the bi-temporal images. Ultimately, the features derived from the perception frame are input into a decoder, which generates a probability map or caption indicating the changes. Change3D can tackle the challenges of bi-temporal image-based methods stated above. First, the distribution of model parameters in Change3D primarily focuses on video feature encoding, a task-specific process adept at extracting changes effectively. Second, intricate change extractors can be eliminated, offering a unified framework for various change detection and captioning tasks.

Extensive experiments on four change detection and captioning tasks (*i.e.*, binary change detection, semantic change detection, building damage assessment, and change captioning) and eight widely recognized datasets (*i.e.*, LEVIR-CD [6], WHU-CD [34], CLCD [49], HRSCD [14], SECOND [81], xBD [28], LEVIR-CC [47], and DUBAI-CC [30]) show that Change3D achieves state-of-the-art performance across the board. Meanwhile, we employ various video models including CNN-based (*e.g.*, I3D [3], SlowFast [22], X3D [21]), and Transformer-based (*e.g.*, UniFormer [40]) with different pre-trained weights (*e.g.*, AVA [27], Charades [65], Something-Something V2 [26], and Kinetics-400 [37]) to confirm the efficacy of the proposed paradigm. Notably, despite the simplicity of the proposed method compared to advanced 2D models, Change3D outperforms methods that utilize these complex structural designs, showcasing the effectiveness of the proposed paradigm for bi-temporal change detection and captioning tasks.

The key contributions of this paper are summarized as follows:

- We revisit the bi-temporal change detection and captioning tasks from a video modeling perspective and

introduce a paradigm called Change3D, a simple yet effective 3D modeling approach for bi-temporal change detection and captioning.
- By injecting the perception frame into the inputs, Change3D efficiently extracts changes using 3D models without the need for elaborate structural designs or complex frameworks.
- Change3D achieves state-of-the-art performance with an efficient video model (*i.e.*, X3D-L) across multiple benchmarks, demonstrating the superiority of the proposed method. Moreover, both quantitative and qualitative analyses validate the efficacy of the new paradigm we have introduced, further solidifying the effectiveness of our approach.

2. Related Work

Change Detection & Captioning. CNN-based approaches have been the mainstream framework in the literature [4, 12, 13, 15, 20, 23–25, 44, 47, 51, 66, 85] for a long time, known for their hierarchical feature modeling capabilities. These works primarily focus on multi-scale feature extraction, difference modeling, foreground-background class imbalance, *etc*. For instance, methods in [12–14] utilize fully convolutional networks to capture hierarchical features for multi-scale feature representations. For adequate differential feature modeling, approaches in [4, 15, 20, 23, 24, 44, 46, 66, 72] incorporate the attention mechanism [70] or feature exchange to establish relational dependencies between bi-temporal features. Several studies [84, 85] address the significant challenge posed by foreground-background class imbalance by developing innovative loss functions. Motivated by the achievements of Vision Transformer [17] and its variants [52, 53, 75] for various visual tasks [10, 42, 80, 82], several works [1, 7, 8, 16, 36, 48, 50, 56, 83, 88] have explored the application of transformers in change detection and captioning tasks. Some of these methods [1, 8, 11, 83, 90] utilize pure transformers, while others [7, 16, 36, 50, 56, 88] adopt CNN-Transformer hybrid architectures. Methods in [1, 8, 11, 83] introduce hierarchical transformer networks, which are both based on the swin transformer [52]. The others typically follow a paradigm in which features extracted by CNN serve as semantic tokens, followed by contextual relation modeling between bi-temporal tokens using transformer.

In contrast to the above, our proposed method considers the change detection and captioning tasks as a video understanding process. It integrates learnable perception frames to efficiently capture changes utilizing a video encoder, thus obviating the necessity for meticulously designed change extractors and providing a unified framework.

Video Understanding. Video understanding in action recognition [21, 22, 32] and detection [18, 40, 43, 54] involves the analysis of video data to identify and localize actions performed by objects within the video frames. Over the past years, video understanding advances have mostly been driven by 3D convolutional neural networks (CNNs) [22, 33, 67]. However, they suffer from difficult optimization and large computation cost problems. To resolve the issue, I3D [3] inflates the pre-trained 2D convolution kernels for better optimization. Methods in [21, 62, 68, 69, 71] try to factorize 3D convolution kernel in different dimensions to reduce complexity, others [35, 41, 45, 73] introduce well-designed temporal modeling modules to enhance the temporal modeling ability for 2D CNNs, meanwhile saving the computation cost. Recently, several works [2, 18, 32, 40, 43, 54] have explored the potential of Vision Transformer [17] for spatiotemporal learning, verifying the outstanding ability to capture long-term dependencies.

Building upon previous studies, we reconceptualize bi-temporal images as comprising two frames akin to a tiny video. Through the integration of perception frames, employing a video encoder facilitates efficient capture of changes.

3. Method

The overall architectures of Change3D are illustrated in Fig. 3, comprising two main components: (1) a video encoder for extracting perception features, and (2) one or more decoders tasked with transforming perception features into change maps or captions. In the following, we first introduce the problem formulation in Sec. 3.1, followed by a detailed exposition of our method in Sec. 3.2.

3.1. Problem Formulation

In this paper, we focus on two key tasks within the remote sensing field: change detection, which includes binary change detection (BCD), semantic change detection (SCD), and building damage assessment (BDA); and change captioning (CC). These tasks are formulated as follows:

Change Detection. Change detection involves identifying and analyzing changes between bi-temporal images captured at different times. Given two images $I_1, I_2 \in \mathcal{R}^{H\times W\times C}$, the change detection process can be formulated as: $(\mathcal{M}_{\text{binary}}, \mathcal{M}_{\text{semantic}}, \mathcal{M}_{\text{damage}}) = \mathcal{F}_{\text{CD}}(I_1, I_2)$. Here, $\mathcal{F}_{\text{CD}}$ is a change detector, $\mathcal{M}_{\text{binary}} \in \{0,1\}^{H\times W}$ represents the binary change map indicating change or no change, $\mathcal{M}_{\text{semantic}} \in \{0,1,\cdots,N_{\text{semantic}}-1\}^{H\times W}$ is the semantic change map classifying land cover into N_{semantic} categories, and $\mathcal{M}_{\text{damage}} \in \{0,1,\cdots,N_{\text{damage}}-1\}^{H\times W}$ denotes the building damage map with N_{damage} levels of damage.

Change Captioning. Change captioning involves generating natural language descriptions that explain differences between bi-temporal images. Given a pair of bi-temporal images $I_1, I_2 \in \mathcal{R}^{H\times W\times C}$, and a vocabulary set $\mathcal{V}$, change captioning process is expressed as: $\mathcal{W} = \mathcal{F}_{\text{CC}}(I_1, I_2) \in \mathcal{V}^L$. Here, $\mathcal{F}_{\text{CC}}(\cdot)$ is a change captioner, and $\mathcal{W}$ represents

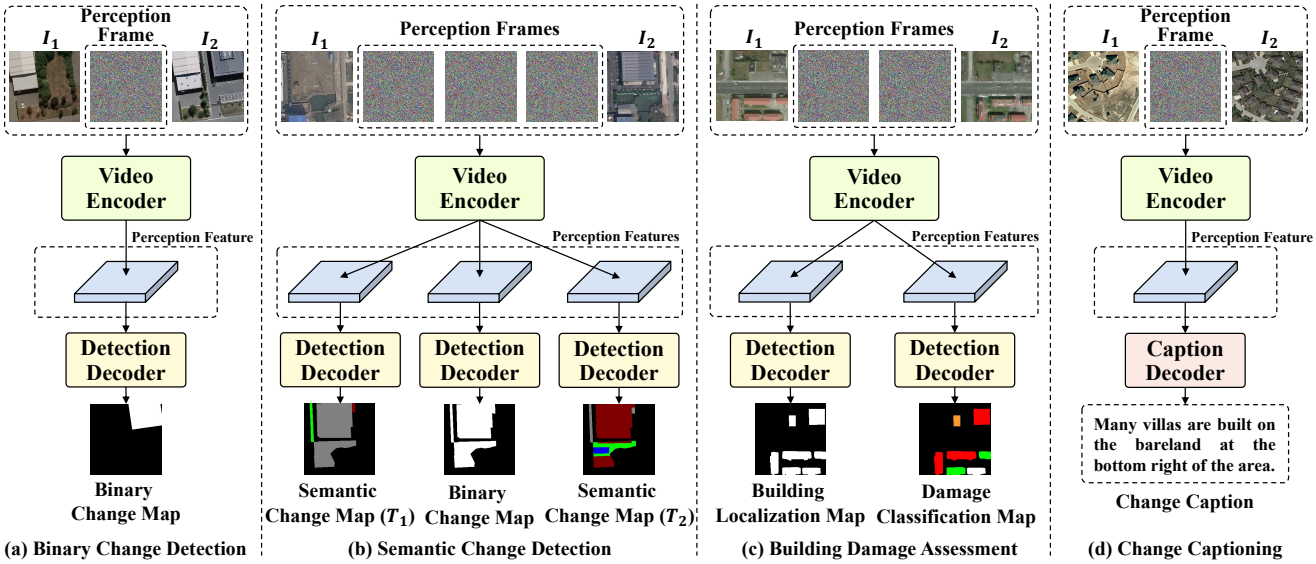


Figure 3. **Overall architectures of Change3D for Binary Change Detection, Semantic Change Detection, Building Damage Assessment, and Change Captioning.** (a) Binary change detection necessitates acquiring a feature to represent changed targets, thus a perception frame is incorporated for sensing. (b) Semantic change detection involves representing semantic changes in T_1 and T_2 alongside binary changes. To accomplish this, three perception frames are integrated to facilitate semantic change learning. (c) Building damage assessment entails expressing two perception features for building localization and damage classification. Therefore, two perception frames are inserted to capture building damage. (d) Change captioning involves generating a feature that represents the altered content, thus incorporating a perception frame for interpreting content changes.

the sequence of words describing the changes, with L being the length of the caption.

3.2. Change3D

To explore the versatility of our proposed method, we apply Change3D to change detection and captioning tasks. As depicted in Fig. 3, we first initialize a set number of perception frames corresponding to the number of decoders for each task. Subsequently, these frames are stacked with bi-temporal images along temporal dimensions to compose the input video frames. Then, a video encoder is employed to facilitate the interaction between the perception frames and bi-temporal images to produce perception features. Finally, the corresponding decoders convert the perception features into change maps or captions.

3.2.1 Perception Feature Extraction

Given a pair of bi-temporal images I_1, I_2, along with learnable perception frames $I_P = \{I_P^1, \cdots, I_P^K\}$, each with dimensions $(H, W, 3)$, where K denotes the length of perception frames, with possible values of 1, 3, 2, or 1, the values corresponding to BCD, SCD, BDA, and CC, respectively. The perception frame functions similarly to the CLS token in Vision Transformers [17], capturing essential information across the whole frame sequence. These frames are concatenated along the temporal dimension to construct a video. Then, a video encoder is employed to extract multi-layer features f, represented as Eq. (1):

$$f = \mathcal{F}_{\text{enc}}(I_1 \copyright I_P \copyright I_2), \tag{1}$$

where $\mathcal{F}_{\text{enc}}(\cdot)$ is a video encoder, © denotes concatenation along temporal dimension, $f \in \{\mathcal{R}^{T \times C_i \times \frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}}}\}_{i=0}^{3}$, where $T = K + 2$ and C_i is the channel dimension. f_i^t indicates the extracted feature of the t-th frame from the i-th encoder layer. We use the multi-layer perception features to detect changes with diverse sizes, denoted as $p_{\text{det}}^{i,j} \in \{f_i^1, \cdots, f_i^{T-2}\}$, where $p_{\text{det}}^{i,j}$ denotes the j-th perception feature in the i-th layer. And we leverage high-level perception feature f_3^1 to caption changes, denoted as p_{cap}.

To augment the representation capacity of perception features, a convolutional layer with the kernel size of 1×1 followed by a ReLU is utilized to integrate the differential features of bi-temporal features f_i^0 and f_i^{T-1} into them.

3.2.2 Decoder and Optimization

Change Decoder. Existing methods [4, 9, 11, 24, 50, 60] utilize complex approaches for capturing bi-temporal changes and predicting change maps. To more convincingly highlight the learning prowess of Change3D, we choose to implement a simpler decoder. Specifically, we employ a cascade convolutional layer followed by an upsampling operation to progressively aggregate perception features from

deep to shallow layers, ultimately restoring them to the original resolution of $H \times W$, as represented in Eq. (2):

$$p_{\text{det}}^{i,j} \leftarrow \text{Deconv}_{4\times4}(\text{Conv}_{1\times1}(p_{\text{det}}^{i+1,j})) + p_{\text{det}}^{i,j}, \quad (2)$$

where $\text{Conv}_{1\times1}$ is a 2D convolution with a kernel size of 1×1, $\text{Deconv}_{4\times4}$ denotes transposed 2D convolution with a kernel size of 4×4 and stride of 2×2.

Finally, a classification layer is applied to transform the shallowest features $p_{\text{det}}^{0,j}$ into change maps $\mathcal{M}$, which is formulated as Eq. (3):

$$\mathcal{M} = \arg\max(\text{Conv}_{3\times3}(p_{\text{det}}^{0,j})), \quad (3)$$

where $\text{Conv}_{3\times3}$ is a 2D convolution with the kernel size of 3×3, $\mathcal{M} \in \{0, \cdots, N-1\}^{H\times W}$, N denotes the number of land cover categories or damage levels, and $\arg\max$ operation is performed along channel dimension.

Caption Decoder. Following common practice, as adopted by [47, 48, 89], we employ a transformer-based [17] decoder to describe the changes. Specifically, the input words w are initially passed through a word embedding layer along with a learnable positional embedding to obtain feature embeddings $e_w \in \mathcal{R}^{L\times D}$, where L represents the word length and D denotes the embedding dimension. Next, masked self-attention and cross-attention blocks are applied to model the changes based on the perception features, as described in Eq. (4):

$$e_w^{'} = \text{CrossAttn}(\text{SelfAttn}(e_w), p_{\text{cap}}) + e_w, \quad (4)$$

where the masking operation is performed on the attention matrix to prevent the leakage of words at subsequent positions. The deepest feature p_{cap} serves as keys and values, while the feature embeddings e_w, enhanced by SelfAttn($\cdot$), function as queries.

Finally, a fully connected (FC) layer followed by a softmax function is applied to $e_w^{'}$ to predict the probability of change captions, which is represented as Eq. (5):

$$\mathcal{W} = \text{softmax}(\text{FC}(e_w^{'})). \quad (5)$$

Similar to existing methods [44, 48, 85, 87], we employ joint loss functions to optimize four tasks. Formulations of the loss functions are as follows:

$$\begin{aligned} \mathcal{L}_{\text{BCD}} &= \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{dice}}, \ \mathcal{L}_{\text{SCD}} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{dice}} + \mathcal{L}_{\text{sim}}, \\ \mathcal{L}_{\text{BDA}} &= \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{dice}}, \ \mathcal{L}_{\text{CC}} = \mathcal{L}_{\text{ce}}, \end{aligned} \quad (6)$$

where $\mathcal{L}_{\text{ce}}$, $\mathcal{L}_{\text{dice}}$, and $\mathcal{L}_{\text{sim}}$ denote cross-entropy, dice [57], and cosine similarity [15] loss, respectively.

4. Experiments

4.1. Experimental Settings

Datasets and Evaluation Metrics: We conduct extensive experiments on eight public datasets: LEVIR-CD [6], WHU-CD [34], CLCD [49], HRSCD [14], SECOND [81], xBD [28], LEVIR-CC [47], and DUBAI-CC [30]. The LEVIR-CD, WHU-CD, and CLCD are three binary change detection datasets, including changes in buildings, roads, and lakes. The HRSCD and SECOND are collected for semantic change detection, including five and six land-cover categories, respectively, for semantic learning. The xBD is a building damage assessment dataset that provides information on four damage categories. LEVIR-CC and DUBAI-CC are change captioning datasets that provide detailed descriptions of building and urbanization changes, with five sentences per image pair. We report change detection performance using F1 score, intersection over union (IoU), overall accuracy (OA), and SeK coefficient, and assess change captioning performance using BLEU-N (N=1,2,3,4), METEOR, ROUGE-L, and CIDEr.

Implementation Details: We adopt I3D [3], Slow-R50 [22], UniFormer-XS [40], and X3D-L [21] as video encoders to verify the effectiveness of our method. Unless otherwise stated, we use X3D-L for our experiments, and we conduct diagnostic studies on the largest publicly available open-source dataset, xBD [28]. All video encoders are pre-trained on Kinetics-400 [37], with the decoder and perception frames initialized randomly. Additionally, we explore the initialization weights pre-trained on other video datasets in Tab. 6. Models are trained using the PyTorch [59] framework on an NVIDIA GeForce RTX 3090 GPU. During training, we apply data augmentation through random resizing, flipping, and cropping. For optimization, we opt for the Adam optimizer [39], with beta values set to (0.9, 0.99) and a weight decay of 1e-4. Initially, the learning rate is 2e-4 and gradually reduces according to a scheduled reduction formula: (1-(curr_iter/max_iter))$^{\alpha}$ $\times$ lr, where α is set to 0.9, and max_iter is set to 200K iterations for xBD, and 80K iterations for the rest datasets. The images of all datasets are cropped into 256$\times$256 non-overlap patches, with a batch size of 8 for HRSCD and SECOND, 32 for LEVIR-CC and DUBAI-CC, and 16 for the others.

4.2. Comparison with State-of-the-Art Methods

Tab. 1-4 represent comparisons to state-of-the-art methods on change detection and captioning tasks across eight datasets. Notably, all the compared methods employ a shared-weight image encoder with a change extractor for feature extraction. Specifically, BIT, ICIFNet, Changer, DMINet, GASNet, AMTNet, Bi-SRNet, DEFO, JFRNet, ChangeOS, PCDASNet, RSICCformer, and SEN apply ResNet [29] or its variants [31] as image encoder, SNUNet and EATDer apply UNet [64] and stacked non-local blocks [76], respectively. In contrast, our approach employs a video encoder to directly capture changes, eliminating the need for specialized change extractors and providing a unified framework for multiple change detection and caption-

Table 1. Performance comparison of different binary change detection methods on LEVIR-CD, WHU-CD, and CLCD datasets. † denotes the results with our reimplementation. The best results are **bolded** and the second-best results are underlined. Values in parentheses indicate the percentage relative to the SOTA method (AMTNet). All results of the three evaluation metrics are described as percentages (%).

Method	#Params(M)	FLOPs(G)	Inference (s/sample)	LEVIR-CD			WHU-CD			CLCD		
				F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
DTCDSCN [51]	41.07 (250%)	20.44 (83%)	0.029	87.43	77.67	98.75	79.92	66.56	98.05	57.47	40.81	94.59
SNUNet [19]	12.04 (73%)	54.82 (222%)	0.036	88.16	78.83	98.82	83.22	71.26	98.44	60.82	43.63	94.90
ChangeFormer [1]	41.03 (250%)	202.79 (822%)	0.081	90.40	82.48	99.04	87.39	77.61	99.11	61.31	44.29	94.98
BIT [7]	3.55 (22%)	10.63 (43%)	0.027	89.31	80.68	98.92	83.98	72.39	98.52	59.93	42.12	94.77
ICIFNet [23]	23.82 (145%)	25.36 (103%)	0.049	89.96	81.75	98.99	88.32	79.24	98.96	68.66	52.27	95.77
Changer † [20]	11.39 (69%)	11.86 (48%)	0.025	90.70	82.99	99.08	89.18	80.48	99.17	67.07	50.46	95.69
DMINet [24]	6.24 (38%)	14.42 (58%)	0.036	90.71	82.99	99.07	88.69	79.68	98.97	67.24	50.65	95.21
GASNet † [85]	23.59 (143%)	23.52 (95%)	0.028	90.52	83.48	99.07	91.75	84.76	99.34	63.84	46.89	94.01
AMTNet [50]	16.44 (100%)	24.67 (100%)	0.032	90.76	83.08	98.96	92.27	85.64	99.32	75.10	60.12	96.45
EATDer † [56]	6.61 (40%)	23.43 (95%)	0.034	91.20	83.80	98.75	90.01	81.97	98.58	72.01	56.19	96.11
Change3D	**1.54** (9%)	**8.29** (34%)	**0.015**	**91.82**	**84.87**	**99.17**	**94.56**	**89.69**	**99.57**	**78.03**	**63.97**	**96.87**

Table 2. Performance comparison of different semantic change detection methods on HRSCD and SECOND datasets. † denotes the results with our reimplementation. The best results are **bolded** and the second-best results are underlined. Values in parentheses indicate the percentage relative to the SOTA method (Bi-SRNet). All results of the four evaluation metrics are described as percentages (%).

Method	#Params(M)	FLOPs(G)	Inference (s/sample)	HRSCD				SECOND			
				F1	mIoU	OA	SeK	F1	mIoU	OA	SeK
HRSCD-S4 [14]	13.71 (59%)	43.69 (23%)	0.024	69.39	67.79	81.32	22.50	58.21	71.15	86.62	18.80
ChangeMask † [88]	8.73 (37%)	37.16 (20%)	0.021	70.59	67.56	81.65	23.43	59.74	71.46	86.93	19.50
SCDNet [60]	39.62 (169%)	116.98 (62%)	0.032	70.80	67.43	81.46	23.61	60.01	70.97	87.40	19.73
SSCD-L † [15]	23.39 (100%)	189.57 (100%)	0.029	69.69	66.21	81.05	21.88	61.22	72.60	87.19	21.86
Bi-SRNet † [15]	23.39 (100%)	189.91 (100%)	0.031	71.72	67.83	82.06	24.59	61.85	72.08	87.20	21.36
MTSCD [11]	94.62 (405%)	290.28 (153%)	0.028	67.56	63.71	72.06	9.03	60.23	71.68	87.04	20.57
JFRNet [4]	23.29 (100%)	47.38 (25%)	0.035	66.65	64.65	76.20	18.78	62.63	72.82	87.10	22.56
DEFO † [44]	26.02 (111%)	401.09 (211%)	0.025	66.49	65.67	81.36	19.20	61.18	72.39	87.01	21.08
Change3D	**1.66** (7%)	**15.19** (8%)	**0.018**	**73.29**	**68.67**	**82.57**	**26.85**	**62.83**	**72.95**	**87.42**	**22.98**

ing tasks. From Tab. 1-4, we can summarize the following key findings: (1) Change3D achieves superior or competitive performance compared to the existing methods across all datasets and evaluation metrics, which demonstrates its effectiveness. (2) Compared with state-of-the-art methods, our proposed method requires ∼6%-13% fewer parameters, ∼8%-34% FLOPs, and has the fastest inference speed, while obtaining comparable performance over all datasets. For example, when compared with AMTNet, which introduces well-designed attention modules for bi-temporal feature interaction, our proposed method achieves significantly better results with little computation cost (9%) and fewer parameters (34%), indicating the efficiency of Change3D. (3) Our proposed method is adaptable to BCD, SCD, BDA, and CC without necessitating architectural redesign, while also achieving superior performance. This demonstrates the generalizability of Change3D across diverse change detection and captioning tasks.

4.3. Diagnostic Study

Effectiveness with different architectures. To investigate the effectiveness of the proposed Change3D with different 3D architectures, we conduct experiments using both CNN-based (*i.e.*, I3D [3], Slow-R50 [22], and X3D-L [21]) and Transformer-based (*i.e.*, UniFormer-XS [40]) models. Key observations from Tab. 5 include: (1) When integrated with our proposed method, all video models exhibit satisfactory performance, indicating the efficacy of Change3D regardless of model architectures. (2) The X3D-L outperforms all other models, possibly due to its use of expansion operations, which perform network expansion across various dimensions (temporal duration, spatial resolution, width, and depth), thus enhancing its inter-frame modeling capacity.

Impact of pre-trained weights. As illustrated in Tab. 6, we explore the impact of pre-trained weights on Change3D by employing diverse initialization methods for initializing the video encoder (defaulting to Slow-R50). These methods encompass random initialization and several publicly accessible pre-training weights within the action recognition domain, *i.e.*, AVA [27], Charades [65], Something-Something V2 (SSv2) [26], and Kinetics-400 (K400) [37]. Key observations from the results are as follows: (1) Without utilizing pre-trained weights, Change3D exhibits inferior performance compared to others, highlighting the effectiveness and necessity of pre-trained weights. (2) Despite being trained on action recognition data, all initialized weights

Table 3. Performance comparison of different building damage assessment methods on xBD dataset. The best results are **bolded** and the second-best results are underlined. Values in parentheses indicate the percentage relative to the SOTA method (PCDASNet). All results of the seven evaluation metrics are described as percentages (%).

Method	#Params(M)	FLOPs(G)	Inference (s/sample)	F_1^{loc}	F_1^{cls}	$F_1^{overall}$	Damage F_1 Per-class			
							Non	Minor	Major	Destroy
xBD baseline [28]	25.72 (99%)	23.15 (24%)	0.021	80.47	3.42	26.54	66.31	14.35	0.94	46.57
Weber et al. [77]	48.29 (186%)	37.48 (39%)	0.029	83.60	70.02	74.10	90.60	49.30	72.20	83.70
ChangeOS-R18 [87]	15.99 (62%)	69.65 (73%)	0.019	84.62	69.87	74.30	88.61	52.10	70.36	79.65
ChangeOS-R34 [87]	26.10 (100%)	74.50 (78%)	0.021	85.16	70.28	74.74	88.63	52.38	71.16	80.08
ChangeOS-R50 [87]	53.02 (204%)	101.35 (106%)	0.025	85.41	70.88	75.24	88.98	53.33	71.24	80.60
ChangeOS-R101 [87]	72.01 (277%)	111.10 (116%)	0.031	85.69	71.14	75.50	89.11	53.11	72.44	80.79
DamFormer [8]	31.38 (121%)	220.68 (230%)	0.065	**86.86**	72.81	77.02	89.86	56.78	72.56	80.51
PCDASNet [72]	26.00 (100%)	95.9 (100%)	0.020	85.48	73.83	77.33	90.12	55.67	75.74	83.91
Change3D	**1.60** (6%)	**11.74** (12%)	**0.016**	85.74	**76.71**	**79.42**	**95.08**	**58.70**	**76.50**	**86.76**

Table 4. Performance comparison of different change captioning methods on LEVIR-CC and DUBAI-CC datasets. The best results are **bolded** and the second-best results are underlined. Values in parentheses indicate the percentage relative to the SOTA method (SEN). Abbreviations B, M, R, and C refer to BLEU, METEOR, ROUGE, and CIDEr, respectively.

Method	#Params(M)	FLOPs(G)	Inference (s/sample)	LEVIR-CC							DUBAI-CC						
				B-1	B-2	B-3	B-4	M	R	C	B-1	B-2	B-3	B-4	M	R	C
DUDA [58]	80.31 (201%)	20.28 (84%)	0.014	81.44	72.22	64.24	57.79	37.15	71.04	124.32	58.82	43.59	33.63	25.39	22.05	48.34	62.78
MCCFormer-S [61]	162.55 (407%)	25.09 (104%)	0.015	79.90	70.87	62.80	56.31	36.17	69.46	120.46	52.97	37.02	27.62	22.57	18.64	43.29	53.81
MCCFormer-D [61]	162.55 (407%)	25.09 (104%)	0.016	80.42	70.87	62.86	56.38	36.37	69.32	120.44	64.65	50.45	39.36	29.48	25.09	51.27	66.51
RSICCformer [47]	172.80 (433%)	27.10 (114%)	0.010	84.72	74.96	67.52	62.11	38.80	74.22	132.62	67.92	53.61	41.37	31.28	25.41	51.96	66.54
PromptCC [48]	408.58 (1024%)	19.88 (84%)	0.013	83.66	75.73	69.10	63.54	38.82	73.72	136.44	70.03	58.41	**49.44**	**40.32**	26.48	55.82	85.44
SEN [89]	39.90 (100%)	24.04 (100%)	0.008	85.10	77.05	70.01	64.09	39.59	74.57	136.02	64.12	50.41	40.28	31.01	23.67	48.19	65.15
Change3D	**5.05** (13%)	**2.39** (10%)	**0.007**	**85.81**	**77.81**	**70.57**	**64.38**	**40.03**	**75.12**	**138.29**	**72.25**	**58.68**	47.13	36.80	**27.06**	**56.04**	**86.19**

Table 5. Study the effectiveness of the proposed method with different 3D architectures on the xBD dataset.

Method	F_1^{loc}	F_1^{cls}	$F_1^{overall}$	Damage F_1 Per-class			
				Non	Minor	Major	Destroy
I3D [3]	84.45	74.32	77.36	94.57	55.30	73.87	86.01
Slow-R50 [22]	85.52	74.36	77.71	94.81	54.67	74.03	**87.34**
UniFormer-XS [40]	85.56	76.11	78.94	95.04	57.75	75.41	87.27
X3D-L [21]	**85.74**	**76.71**	**79.42**	**95.08**	**58.70**	**76.50**	86.76

Table 6. Investigation on the impact of different pre-trained weights on the xBD dataset.

Pre-trained	F_1^{loc}	F_1^{cls}	$F_1^{overall}$	Damage F_1 Per-class			
				Non	Minor	Major	Destroy
Random Init	81.00	71.54	74.38	94.11	51.99	71.78	82.53
AVA [27]	85.19	75.41	77.94	94.61	55.94	73.47	86.24
Charades [65]	85.31	75.51	78.05	94.65	56.13	74.02	**86.83**
SSv2 [26]	84.76	74.52	77.89	94.77	55.13	73.37	86.35
K400 [37]	**85.87**	**75.56**	**78.67**	**94.79**	**58.02**	**75.03**	86.30

demonstrate superior performance, indicating the ability of Change3D to bridge the gap between disparate domains and effectively transfer knowledge from action recognition to the field of change detection and captioning. (3) By utilizing pre-training weights from K400, Change3D achieves optimal results. This is mainly due to the K400 dataset providing a larger volume of data and more diverse scenarios compared to the other three datasets, which enhances the representational capability of the pre-trained model.

Impact of perception frame insertion position. As depicted in Tab. 7, this study investigates the effects of the insertion position of perception frames. Particularly, the *sandwiched* perception frame positioned centrally exhibits optimal performance. This optimal performance is primarily attributed to the ability of video encoding to allow perception frames to interact effectively with adjacent bi-temporal images, thereby detecting differences and accurately capturing regions of change.

Impact of different ways obtaining perception features. We conduct four experiments to investigate the impact of various approaches for acquiring perception features. Specifically, FC represents a depth-wise 3D convolution with a kernel size of $3\times1\times1$ and stride of $3\times1\times1$ to aggregate the features. Mean and Max operations refer to average and max pooling along the temporal dimension, respectively. Feature selection involves choosing the perception feature based on the inserted index of the perception frame. Tab. 8 illustrates that the selection methodology yields superior outcomes, effectively capturing changes through the interaction of features between perception frames and bi-temporal images.

Effectiveness of perception frames. We conduct experiments to investigate the effectiveness of integrating perception frames on HRSCD and SECOND datasets. The first line denotes that without incorporating perception frames,

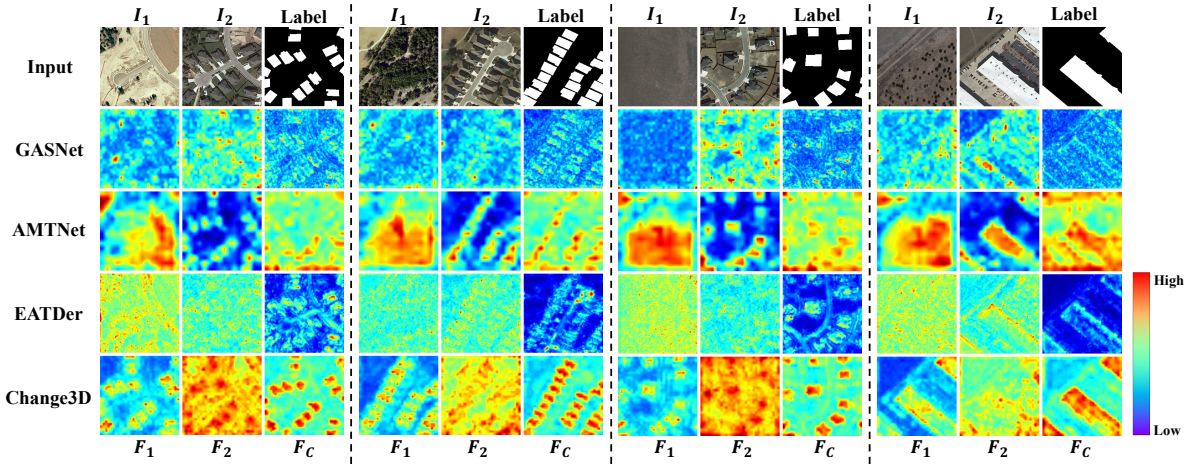


Figure 4. Visualization of bi-temporal features F_1, F_2, and extracted changes F_C. Our method directly focuses on changes during video encoding without intricate change extractors. The color bar on the right indicates the attention distribution for different colors.

Table 7. Investigation on the impact of inserting perception frames I_P^1 and I_P^2 into different positions within the xBD dataset.

Position	F_1^{loc}	F_1^{cls}	$F_1^{overall}$	Damage F_1 Per-class			
				Non	Minor	Major	Destroy
$\{I_P^1, I_P^2, I_1, I_2\}$	85.36	75.67	78.58	94.87	57.58	74.65	85.67
$\{I_1, I_2, I_P^1, I_P^2\}$	85.73	75.45	78.53	95.02	57.53	74.51	85.57
$\{I_1, I_P^1, I_P^2, I_2\}$	**85.74**	**76.71**	**79.42**	**95.08**	**58.70**	**76.50**	**86.76**

Table 8. Investigation on the impact of different ways of obtaining perception features on xBD dataset.

Method	F_1^{loc}	F_1^{cls}	$F_1^{overall}$	Damage F_1 Per-class			
				Non	Minor	Major	Destroy
FC	85.60	75.75	78.79	94.97	57.60	74.86	86.27
Max	85.57	75.30	78.47	94.94	56.63	74.55	86.52
Mean	85.64	74.56	77.88	94.79	55.42	74.68	85.68
Selected	**85.74**	**76.71**	**79.42**	**95.08**	**58.70**	**76.50**	**86.76**

the output from the feature extractor is reshaped into feature maps that correspond to the number of perception features needed for the decoders. The second line represents the inclusion of perception frames. As shown in Tab. 9, performance improves with the integration of perception frames, demonstrating their effectiveness.

Attention visualization. Fig. 4 illustrates attention maps from the last layer during image encoding and change extraction or video encoding, using four randomly selected samples from the LEVIR-CD dataset. Two significant observations emerge from the attention maps: (1) The perception feature of Change3D can efficiently discern changes through interaction with bi-temporal features, eliminating the need for dedicated change extractors. Conversely, other methods inadequately model change information during image feature encoding, underscoring the need for change extractors. (2) Moreover, for the extracted features from bi-temporal images, Change3D distinguishes itself from other methods by effectively concentrating on the changed regions. This capability ensures that the feature extraction process aligns with change detection and captioning tasks.

Table 9. Study the effectiveness of perception frames on two semantic change detection datasets.

Method	HRSCD				SECOND			
	F_{scd}	mIoU	OA	SeK	F_{scd}	mIoU	OA	SeK
Channel to Frames	69.48	66.05	81.34	21.61	59.91	71.62	86.89	19.92
w/ Perception Frames	**73.29**	**68.67**	**82.57**	**26.85**	**62.83**	**72.95**	**87.42**	**22.98**

Please refer to *Supplementary Material* for more experimental details, diagnostic studies, detailed architecture, theoretical analysis of video encoding, and qualitative results.

5. Conclusion

This paper introduces a new paradigm, namely Change3D, to reconsider change detection and captioning from a video modeling perspective. It treats bi-temporal images as composed of two frames, resembling a miniature video. Through the integration of learnable perception frames between bi-temporal images, a video encoder with inter-frame modeling capabilities enables direct interaction between the perception frames and the images, allowing them to discern differences. Furthermore, this approach eliminates the need for complex, task-specific change extractors, providing a unified framework. We demonstrate the effectiveness of Change3D across eight well-recognized datasets with various model architectures. We hope the insight of this paper could inspire further study in more related computer vision tasks, *e.g.*, depth completion and image matting.

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